

PES UNIVERSITY

Department of Computer Science & Engineering



# Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #05 — Academic Elective Bidding &  
Allocation System

**Course:** Software Engineering Lab

## DELIVERABLE 1

## Requirements Table

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**System:** Academic Elective Bidding & Allocation System

**Actors:** Student, Academic Registrar

**Context:** University departments need an automated elective course selection mechanism where students allocate bidding credits across preferences and the engine solves allocation constraints while preventing timetable collisions.

### Functional Requirements (FR-001 to FR-005)

| REQ ID | TYPE       | DESCRIPTION                                                                                                                                                                                             | PRIORITY | ACCEPTANCE CRITERIA                                                                                                                                                                                                            | RATIONALE                                                                                           |
|--------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| FR-001 | Functional | The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.                                                | High     | <b>Pass:</b> Credits deduct accurately and prerequisites validated.<br><b>Fail:</b> Total allocated credits exceed 100 or unmet prerequisites bypassed.                                                                        | Core bidding functionality; ensures fair, merit-based allocation and enforces academic eligibility. |
| FR-002 | Functional | The system shall display a catalogue of available elective courses showing course name, instructor, schedule, available seats, and prerequisite requirements.                                           | High     | <b>Pass:</b> Catalogue loads within 3 seconds and shows accurate, real-time seat counts matching the database.<br><b>Fail:</b> Stale or missing course data displayed, or page fails to render.                                | Students need complete, up-to-date information to make informed bidding decisions.                  |
| FR-003 | Functional | The system shall detect and prevent timetable collisions when a student bids on electives with overlapping class schedules, and display a clear warning identifying the conflicting courses.            | High     | <b>Pass:</b> Submitting bids for two overlapping-schedule courses triggers a collision warning and blocks submission.<br><b>Fail:</b> Overlapping bids are accepted without any alert.                                         | Prevents students from being allocated to courses they cannot physically attend.                    |
| FR-004 | Functional | The system shall execute an allocation engine that processes all submitted student bids, optimises seat assignments based on credit weightings, and publishes final allocation results upon completion. | High     | <b>Pass:</b> After the engine runs, every student receives allocations that respect seat capacity and credit rankings; results are visible in the portal.<br><b>Fail:</b> Over-allocation of seats or unprocessed bids remain. | Central engine that converts bids into fair, constraint-satisfying course assignments.              |

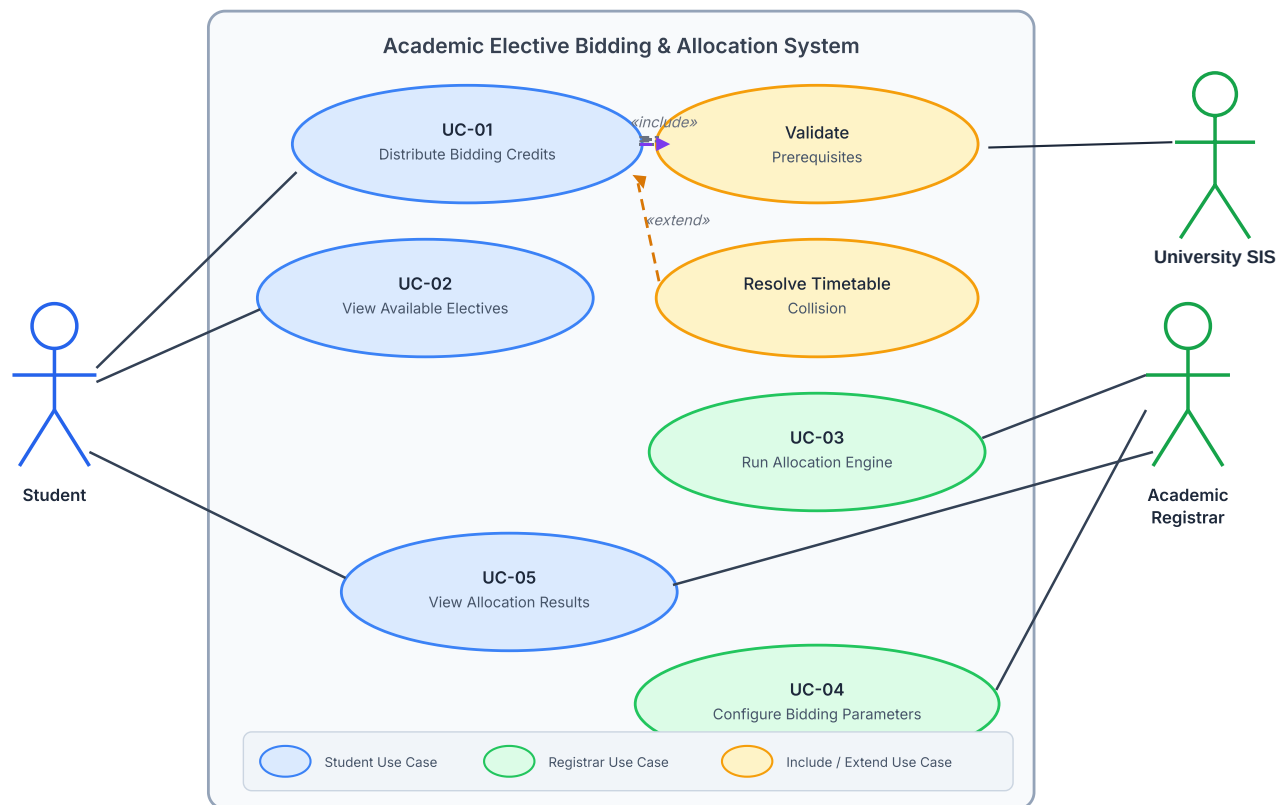
| REQ ID        | TYPE       | DESCRIPTION                                                                                                                                                                                            | PRIORITY | ACCEPTANCE CRITERIA                                                                                                                                                                                                | RATIONALE                                                                                     |
|---------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>FR-005</b> | Functional | The system shall allow the Academic Registrar to configure bidding parameters including bidding window open/close dates, per-course seat capacities, and maximum credits allocable to a single course. | Medium   | <b>Pass:</b> Registrar updates bidding window dates and seat cap; changes reflect immediately in the student-facing interface.<br><b>Fail:</b> Configuration changes are not persisted or not visible to students. | Gives the registrar administrative control to adapt the bidding cycle to institutional needs. |

### Non-Functional Requirements (NFR-001 to NFR-002)

| REQ ID         | TYPE        | DESCRIPTION                                                                                                                                                                                                                 | PRIORITY | ACCEPTANCE CRITERIA                                                                                                                                                                                                                                  | RATIONALE                                                                                            |
|----------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>NFR-001</b> | Performance | The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.                                                                                                   | High     | <b>Pass:</b> Benchmarking tests confirm target latency under simulated peak load of 5,000 concurrent bids.<br><b>Fail:</b> Solver exceeds 30-second threshold or fails under concurrent load.                                                        | Ensures timely results so registration deadlines are met even at scale.                              |
| <b>NFR-002</b> | Security    | The system shall enforce role-based access control so that students can only view and modify their own bids, and registrars have administrative privileges; all data in transit shall be encrypted using TLS 1.2 or higher. | High     | <b>Pass:</b> Penetration test confirms no cross-user data leakage; TLS certificate validation passes; unauthorised API calls return HTTP 403.<br><b>Fail:</b> A student can view/modify another student's bids, or data is transmitted in plaintext. | Protects sensitive academic and personal data and complies with university data-governance policies. |

## DELIVERABLE 2

## UML Use-Case Diagram



### Diagram Summary

| ELEMENT | NAME                         | ACTOR(S)                    | RELATIONSHIP         |
|---------|------------------------------|-----------------------------|----------------------|
| UC-01   | Distribute Bidding Credits   | Student                     | —                    |
| UC-02   | View Available Electives     | Student                     | —                    |
| UC-03   | Run Allocation Engine        | Academic Registrar          | —                    |
| UC-04   | Configure Bidding Parameters | Academic Registrar          | —                    |
| UC-05   | View Allocation Results      | Student, Academic Registrar | —                    |
| —       | Validate Prerequisites       | University SIS              | «include» from UC-01 |
| —       | Resolve Timetable Collision  | —                           | «extend» to UC-01    |

# DELIVERABLE 3

## Use-Case Flow Specification

|               |         |               |                                                                                                                                                                                |
|---------------|---------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID   | UC-01   | Use Case Name | Distribute Bidding Credits                                                                                                                                                     |
| Primary Actor | Student | Description   | A student distributes 100 bidding credits across ranked elective preferences. The system validates prerequisites and checks timetable collisions before confirming submission. |

**Preconditions**

- Student is authenticated and logged into the Elective Bidding Portal.
- Bidding window is open within the Registrar-configured period.
- Student has not finalised a bid for this cycle (Draft/Not Started).
- Elective catalogue is published with at least one available course.

**Postconditions**

- Credit allocations are recorded against selected electives.
- Total allocated credits equal exactly 100.
- Prerequisites are validated as satisfied.
- No selected courses overlap in timetable slots.
- Bid status is Submitted and confirmation is displayed.

### Main Success Scenario

|    |         |                                                                                                      |
|----|---------|------------------------------------------------------------------------------------------------------|
| 1  | Student | Navigates to the "Elective Bidding" page from the student dashboard.                                 |
| 2  | System  | Displays available electives with course name, instructor, schedule, seats, and prerequisites.       |
| 3  | Student | Reviews the catalogue and selects up to 6 elective preferences.                                      |
| 4  | System  | Validates prerequisites for each selected course. («include» Validate Prerequisites)                 |
| 5  | System  | All prerequisite checks pass; displays the credit allocation form.                                   |
| 6  | Student | Distributes 100 bidding credits across selected courses.                                             |
| 7  | System  | Validates total credits = 100 and each course is within the Registrar's per-course credit limit.     |
| 8  | System  | Checks all selected course schedules for overlapping time slots.                                     |
| 9  | System  | No collisions detected; displays confirmation summary with courses, credits, and schedules.          |
| 10 | Student | Reviews the summary and clicks "Submit Bid."                                                         |
| 11 | System  | Persists the bid, sets status to Submitted, and displays a confirmation with a bid reference number. |

### Alternate Flows

**AF-1: Prerequisite Not Met (branches from Step 4)**  
4a.1 System detects unmet prerequisite. 4a.2 Displays a warning identifying the prerequisite/course. 4a.3 Student removes or replaces the ineligible course. 4a.4 Flow resumes at Step 4.

**AF-2: Credits Do Not Sum to 100 (branches from Step 7)**  
7a.1 System detects total ≠ 100. 7a.2 Displays the current total and correction message. 7a.3 Student adjusts credit distribution. 7a.4 Flow resumes at Step 7.

**AF-3: Timetable Collision Detected (branches from Step 8)**  
8a.1 System detects overlapping schedules. («extend» Resolve Timetable Collision) 8a.2 Displays the conflicting courses and time. 8a.3 Student removes/replaces a conflicting course. 8a.4 Flow resumes at Step 4 for re-validation.